22GE002 & COMPUTATIONA PROBLEM SOLVING

1. *Limited search capability - The system supports only basic searches, so finding specific or detailed student information is difficult

*Duplicate records - Without validation, the same student data may be entered multiple times causing ~~redundancy and~~ redundancy and confusion

*Poor data management - Updating and maintaining records becomes difficult, reducing accuracy and efficiency of the database system.

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2. *A flat-file database is suitable because the library stores only a small amount of simple data in one file

*It is easy to create, maintain and does not require relationships or links between records.

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3. * i) Public cloud - Used by the startup company.

ii) Reason: Low cost, scalable and managed by providers like AWS

* i) Private cloud - Used by the bank

ii) Reason: Provides high security and dedicated storage for sensitive data

* i) Hybrid cloud - Used by the multinational company

ii) Reason: combines private cloud for sensitive data and public cloud for scalable general data storage

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4. 1) Amazon S3 (AWS) - Suitable for the media company

Reason: Provides scalable and reliable storage for large video files & backups

2) Azure storage - Suitable for enterprise organisations

Reason: Offers secure storage, collaboration tools and compliance features for enterprise use

5. * i) Sales records in tables - structured ata
* ii) Reason: Data is organized in rows and columns with fixed format
* i) Web data in XML format - semi-structured data
ii) Reason: XML has tags and hierachy but no fixed table structure
* i) Emails and images - unstructured data
ii) Reason : Data does not follow a predefined format or structure

6. * Data type : Flat file database
* key characteristic : Data is stored in a single table/file without relationships between tables

7. * The employee logs into the cloud storage platform using internet.
* The required project file is searched and retrieved from cloud storage.
* The file is opened/downloaded securely for further processing and use

8. * Accessibility - Data can be accessed from different devices and locations through internet
* Backup and Recovery - Important files can be recovered easily during system failures, ensuring data safety

9. a) selection operation (σ DeptName = "MECH")
HOD of MECH department = Dr. Suresh
b) Projection Operation (π Name)

<u>Name</u>
Ravi
Meenu
Arjun
Kiran
Divya
Suresh

c) Set operations on Dept Name
Students Table Dept Names : {CSE, ECE, IT, MECH}
Departments Table Dept Names : {CSE, IT, MECH, EEE, CIVIL}

*Union (∪) = {CSE, ECE, IT, MECH, EEE, CIVIL}
*Intersection (∩) = {CSE, IT, MECH}
*Difference (Students - Departments = {ECE}

10.
a) DROP Phone column:

ALTER TABLE Employee DROP COLUMN Phone;

Output:

Eid	Ename	Dept	Email_id	Salary
101	Adams	Reseach	adams@gmail.com	50000.00
102	Allen	Sales	allen@gmail.com	15000.50
⋮	⋮	⋮	⋮	⋮

similary write all rows without phone column

b) Modify Salary Datatype to integer:

ALTER TABLE Employee MODIFY Salary INT;

Output:

Eid	Ename	Dept	Email_id	Phone	Salary
101	Adams	Research	adams@gmail.com	11111	50000
102	Allen	Sales	allen@gmail.com	22222	15000
⋮	⋮	⋮	⋮	⋮	⋮

similary write all rows without decimal points in salary column

11. * i) Entity Integrity - PatianID is used as the primary key and cannot be NULL or duplicate
ii) impact : Ensures each patient record is uniquely identified

* i) Domain Integrity - Attributes like Age are restrict to valid values using data types and constraints
ii) Impact: Prevents invalid data entry

* i) Referential integrity - Doctor details linked with patients through foreign keys cannot be deleted unnecessarily
ii) impact: Maintains consistency between related tables

* i) User-Defined Integrity - Insurance eligibility rules are defined according to hospital requirements
ii) Impact : Ensures database follows organizational policies
* overall impact : Improves accuracy, consistency and reliability of the hospital database system

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12. i) * CREATE - Create a new table
* eg: CREATE students (
StudentID INT,
Name VARCHAR(50)
);
ii) * ALTER - Modifies table structure
* eg: ALTER TABLE Student ADD Email VARCHAR(50);
iii) * DROP - Deletes a table permanently
* eg: DROP TABLE Student;
iv) * RENAME - changes table name
* eg: RENAME TABLE Student to Student_Details
v) * ~~Role~~ of DDL - used to create, modify and manage database structures efficiently

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13. i) Web server - Delivers Web pages and Websites to employees through browsers.
ii) Database server - stores and retrives organizational data from a centerlized system.
iii) File server - share files and documents within the office network
iv) Mail server - Handles ~~that~~ sending and receiving of emails in organisation
v) DNS server - Resolves domain names into IP address while browsing websites

14.
a) * INSERT INTO Student (Id, Name, Address, Age, Department, Marks)
VALUES (5, 'Thiyagarajan', 'Salem', 18, 'CSE', 91);

* INSERT INTO Student (Id, Name, Address, Age, Department, Marks)
VALUES (6, 'Suresh', 'Pune', 28, 'AIDS', 96);

b) i) UPDATE Student SET Address = 'Mumbai' WHERE Name = 'Mani';
ii) DELETE FROM Student WHERE Id = 5

15.

RAID Level	Performance	Reliability
RAID 0	High speed	No data protection
RAID 1	Moderate speed	High Reliability
RAID 5	Good performance	can recover from one disk failure
RAID 10	very high speed	very high Reliability

* RAID 10 is most reliable RAID because it provides both high performance and strong data protection

16. 1) GRANT - Used to give access permissions
eg: GRANT SELECT, UPDATE ON Customer to Manager;
2) REVOKE - Used to remove access permissions
eg: REVOKE UPDATE ON Customer FROM Employee;
3) Roles - Helps the DBA maintain secure and controlled database access.

17. * Sender (Laptop) - Sends the video file
* Receiver (Smart TV) - Receives the video file
* Transmission medium (Fibre optical cable) - Transfers data
* Protocol - Ensures data exchange
* Message (video file) - Information being transmitted

18. *Issue: File reached wrong user
~~*Violated~~
*Violated characteristic: Delivery - Data must reach the correct destination

19. *Using 192.168.12.0/26, the four subnet network address are
i) Subnet A : 192.168.12.0
ii) Subnet B : 192.168.12.64
iii) Subnet C : 192.168.12.128
iv) Subnet D : 192.168.12.192

20. Static IP Assignment : Printers and servers because IP addresses are assigned manually
Dynamic IP Assignment : User devices because IP addresses are assigned automatically by a central system (DHCP)

21. *Type of Data flow : Full Duplex
*Reason: Both devices can send and receives data at the same time
*Difference :
- Simplex: One-way communication only
- Half Duplex : Two-way communication, but one at a time
- Full Duplex : Two-way communication simultaneously

22. *Issue; Data was altered during transmission
*Violated characteristic: Accuracy - Data must be received without errors or changes.

23. *For network 192.168.12.0/26 is divided into 4 subnets
*Subnet B Network address : 192.168.12.64
*Subnet C Broadcast address : 192.168.12.191
*IP 192.168.12.200 belongs to : subnet D

24. i) *Permanent devices : static IP Assignment
*In this, IP Address is configured manually
ii) *Temporary devices: Dynamic IP Assignment
*In this, IP address is assigned automatically by DHCD

25. * The network shown is a Mesh Topology where every node is connected to all other nodes.

* Data travels through direct dedicated links for fast communication.

* Multiple connections increase network reliability and performance

* If one link fails, data can use another path, so communication continues

* Mesh topology is highly secure and fault tolerant but costly to install

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26. * Access Layer: connects user devices to network

* Distribution Layer: Routes and manages traffic between LANs / VLANs

* core Layer: High-speed backbone connecting all buildings

* Data flow:

Access Layer → Distribution Layer → core Layer → destination LAN

* Role of Load Balancing:

Distributes traffic evenly to reduce congestion and improve speed and reliability

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27. * Star Topology is best for the office network

* All devices are connected to a central switch / hub

* Easy to add or remove devices without affecting others

* communication is efficient through the central device

* Easy network management and fault tolerance

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28. * The student enters www.camps.edu in the web browser

* The DNS resolver contact the Root server, which directs to the .edu server

* The TLD (Top Level Domain) server gives the address of the authoritative server for campus.edu

~~* The DNS resolver contacts the Root server, w~~

* The authoritative DNS server returns the correct IP address of the website
* Using this IP address, the browser connects to the website successfully.

* Differences:

Forward DNS	Reverse DNS
converts a domain name into an IP address	converts an IP address into a domain name

29. * IPv4 addresses are classified into Class A, B, C, D and E
* Class A - Used for very large networks like multinational companies
* Class B - Suitable for large organisations with many devices
* Class C - Best for small businesses and local networks
* Class D is for multicast and class E is for research purposes

30. * Core Layer provides high-speed backbone communication between campus buildings.
* Distribution Layer manages routing, traffic control and policies between departments
* Access layer connects end devices like computers, printers and WiFi
* The layered design improves scalability, performance and easy management
* It also helps in better security, fault isolation and reliable communication

31. * IP Address : 192.168.10.25

* Subnet Mask : 255.255.255.0

* The subnet mask shows the first 24 bit as the network portion and the last 8 bit as the host portion

* Network Address : 192.168.10.0

* Host Address / Host ID : 0.0.0.25 (host part = 25)

32. * Head office to two branches uses Point-to-Point WAN (dedicated secure link)

* Multiple branch officies use switched WAN (shared network communication)

* Point-to-Point WAN provides secure and reliable communication

* Switched WAN connects many locations efficiently and economically

* Switched WAN is best for global communication because it suppost large-scale connectivity